Olist E-Commerce Analytics

GIT Link : https://github.com/Abineswarii/olist-ecommerce-analytics.git

Business Review | Data Engineering and Analytic

Project objective

Build a Python and PostgreSQL pipeline that loads Olist data into RAW and STAGING tables, creates an order-level analytical mart, checks data quality, and answers business questions with SQL and charts.

North Star and guardrail metrics

Definition: customers with two or more orders divided by all unique customer identities in the mart. This measures repeat behaviour; it is not a direct measure of profitability.

Executive snapshot

Metric	Result	Formula / interpretation
North Star: repeat purchase rate	3.12%	2,997 customers with 2+ orders ÷ 96,096 unique customers.
Late-delivery rate	8.11%	7,827 late orders ÷ 96,476 valid delivery-delay records.
Median delivery delay	−11.95 days	Median(delivered timestamp − estimated timestamp); negative means earlier.
Average review score	4.09 / 5	Average latest retained review; 98,673 orders have a retained review.

Main finding

Orders delivered more than seven days late averaged 1.73/5, compared with 4.32/5 for orders delivered more than seven days early. RJ and BA are proposed for a pilot because they combine meaningful late-order volume with elevated late rates.

Headline insight: Orders more than seven days late averaged 1.73/5, compared with 4.32/5 for orders more than seven days early. The results support investigating delivery reliability, while treating the relationship as an association rather than proof of causation.

Revenue definition: monthly and category revenue are calculated as SUM(order_items.price), excluding freight and payment values. The repeat-purchase query counts orders represented in the mart without filtering by order status

1.Pipeline and data quality

The pipeline loads CSV files into RAW tables, builds cleaned STAGING tables, runs pre-MART quality checks, creates mart_order_summary, and then runs post-MART checks and warnings. The pipeline log records each stage and successful completion. RAW tables are replaced; STAGING and MART tables are rebuilt on each run.

Pipeline verification

Table	Observed rows	Purpose
raw_orders	99,441	Raw order source
stg_orders	99,441	Cleaned order data
mart_order_summary	99,441	One row per order
raw_order_items	112,650	Raw order-item source
stg_order_items	112,650	Cleaned order items

Quality-check results

Check	Failing rows	Status / handling
Duplicate order IDs	0	PASS — no duplicates in staging
Null order IDs	0	PASS — no missing order identifiers
Invalid delivery dates	0	PASS — no delivery date before purchase
Orders without customers	0	PASS — customer relationship validated
Negative item values	0	PASS — no negative price/freight values
Invalid review scores	0	PASS — review scores within 1–5
Orders with multiple reviews	547 orders	WARNING — retain latest review by answer timestamp; review_id breaks ties
Delivered orders missing delivery date	8 orders	WARNING — retain, exclude from delivery-delay metrics
Payment vs item + freight mismatch	1,076 (1.08%)	WARNING — flag in MART, retain rows; BRL 0.01 tolerance

Payment reconciliation: 98,365 of 99,441 orders reconcile within BRL 0.01; 1,076 do not. The total absolute difference is BRL 166,004.63. The MART contains payment_reconciliation_difference and payment_reconciliation_flag. These flags identify differences; they do not prove that a transaction is invalid.

Installment investigation: differences appear across multiple payment types and installment counts. For example, credit-card orders with 10 installments had 94 mismatches among 5,228 orders, while boleto with one installment had 184 among 19,784. This does not prove installments caused the differences.

2. SQL Analysis and Findings

Q1. Monthly merchandise value: CTEs and LAG() calculate month-over-month change. Zero-value months remain visible. Growth is marked N/M when the prior month is zero.

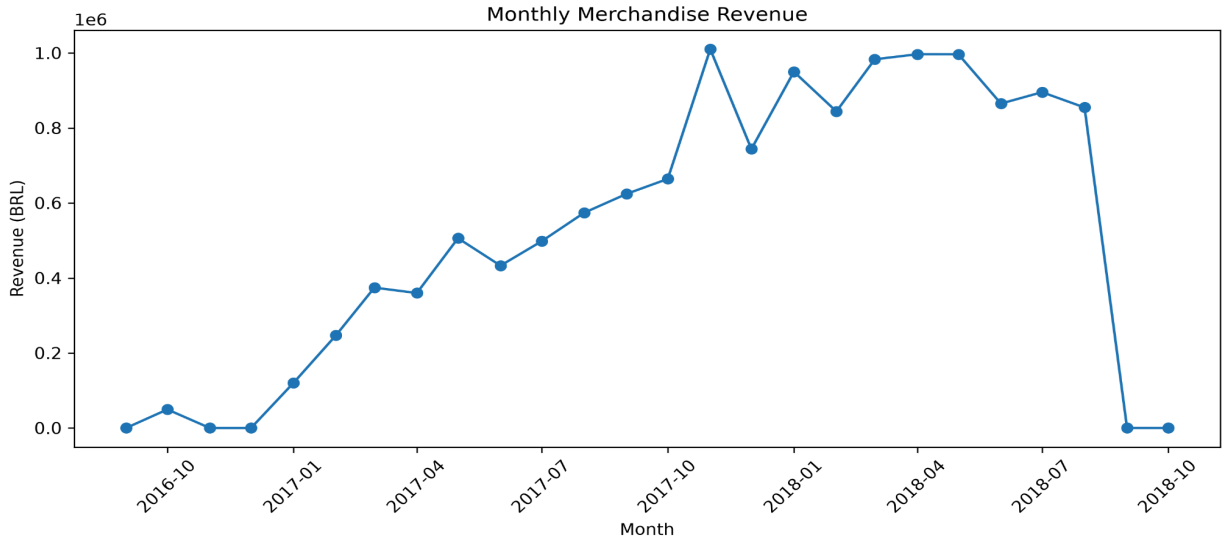


Figure 1. Monthly merchandise value (BRL). Sep/Oct 2018 are sparse tail observations and should not be interpreted as a collapse.

Month	Value (BRL)	Previous (BRL)	MoM change
Sep-2016	267.36	—	—
Oct-2016	49,507.66	267.36	18417.23%
Nov-2016	0.00	49,507.66	-100.00%
Dec-2016	10.90	0.00	N/M (prior = 0)
Jan-2017	120,312.87	10.90	1103687.80%
Feb-2017	247,303.02	120,312.87	105.55%
Mar-2017	374,344.30	247,303.02	51.37%
Apr-2017	359,927.23	374,344.30	-3.85%
May-2017	506,071.14	359,927.23	40.60%
Jun-2017	433,038.60	506,071.14	-14.43%
Jul-2017	498,031.48	433,038.60	15.01%
Aug-2017	573,971.68	498,031.48	15.25%
Sep-2017	624,401.69	573,971.68	8.79%
Month	**Value (BRL)**	**Previous (BRL)**	**MoM change**
Oct-2017	664,219.43	624,401.69	6.38%
Nov-2017	1,010,271.37	664,219.43	52.10%
Dec-2017	743,914.17	1,010,271.37	-26.36%
Jan-2018	950,030.36	743,914.17	27.71%
Feb-2018	844,178.36	950,030.36	-11.14%
Mar-2018	983,213.44	844,178.36	16.47%
Apr-2018	996,647.75	983,213.44	1.37%
May-2018	996,517.68	996,647.75	-0.01%
Jun-2018	865,124.31	996,517.68	-13.19%
Jul-2018	895,507.22	865,124.31	3.51%
Aug-2018	854,686.33	895,507.22	-4.56%
Sep-2018	145.00	854,686.33	-99.98%
Oct-2018	0.00	145.00	-100.00%

Reading the trend: Nov-2017 was the peak month at BRL 1,010,271.37. Dec-2017 fell 26.36%; Jan-2018 rose 27.71%. Dec-2016 compares with Nov-2016, which has zero recorded value.

3. Business Findings and Visuals

Q2. Top three categories by state. DENSE_RANK() ranks categories by merchandise value in each state.

State	Rank	Category	Revenue (BRL)
MG	1	health_beauty	157,558.30
MG	2	bed_bath_table	129,643.98
MG	3	watches_gifts	123,759.23
RJ	1	watches_gifts	185,379.65
RJ	2	bed_bath_table	148,035.23
RJ	3	health_beauty	145,298.62
SP	1	bed_bath_table	478,284.52
SP	2	health_beauty	462,305.22
SP	3	watches_gifts	435,009.92

Q3. Seven-day rolling orders. Missing calendar dates were filled with zero before applying the rolling window.

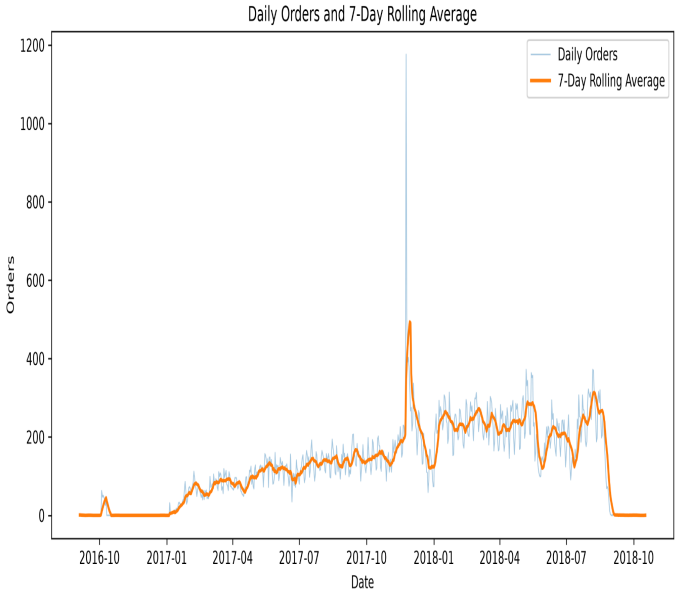


Figure 2. Daily orders and seven-day rolling average.

Q4. Repeat purchase: 2,997 of 96,096 unique customers had more than one order, giving a 3.12% repeat-purchase rate

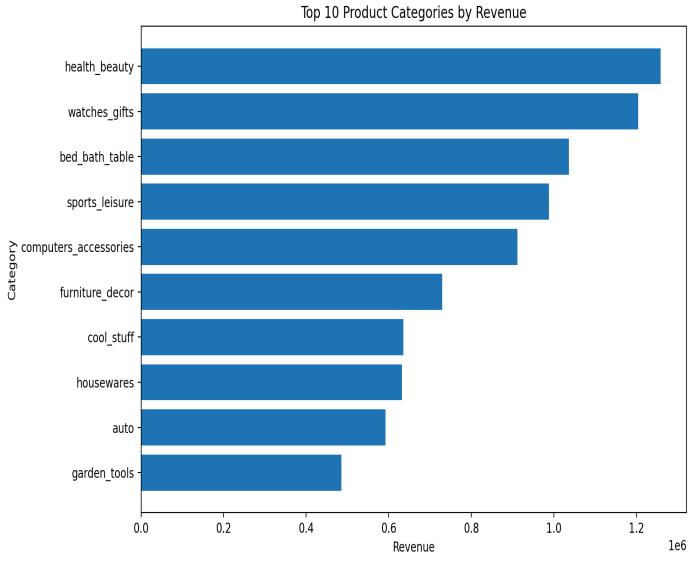


Figure 3. Top10 categories by merchandise value .

4. Delivery Findings

Q5. Delivery delay vs review score. The query uses the latest retained review per order and includes only orders with both a review and valid delivery-delay value.

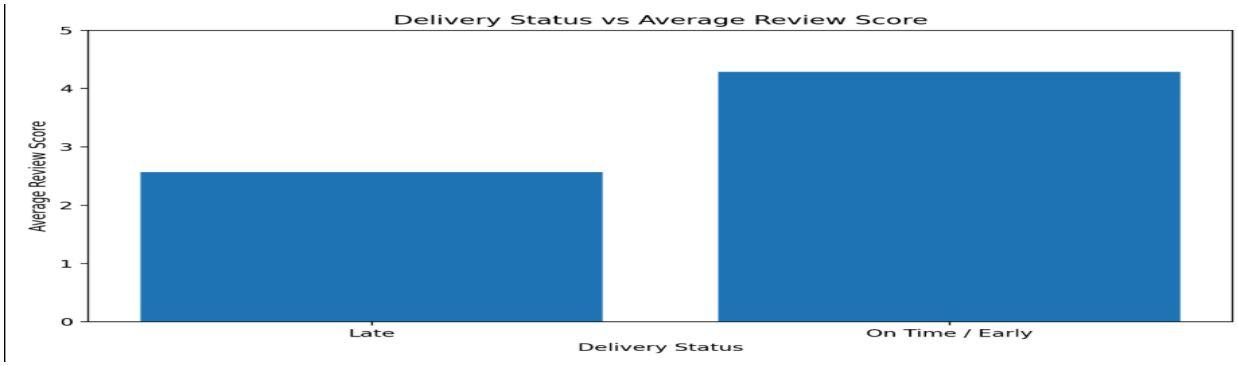


Figure 4. Delivery delay and review score relationship

Memo to the VP

Subject: Improve delivery reliability and investigate payment exceptions

Key finding

Delivery performance is associated with customer review outcomes. Among delivered orders with a review score of 1, 36.79% were late, compared with 1.86% for score-5 orders. Across the dataset, the repeat-purchase rate is 3.00% among customers with at least one delivered order. These results identify opportunities to investigate delivery experience and customer retention.

Recommended actions

1. Introduce a weekly at-risk shipment report using estimated delivery dates and current fulfilment status.
2. Alert the fulfilment team when an order is likely to be late and communicate proactively with affected customers.
3. Track late-delivery rate and review-score trends together each week to evaluate whether interventions improve outcomes.
4. Investigate the 1076 payment reconciliation mismatches and the one delivered order without a payment record before relying on payment totals for financial reporting and validate the repeat-customer cohort and segment by acquisition period, category, and geography before selecting retention campaigns.

Success measures

Monitor weekly delivered order value, delivered order count, average order value, late-delivery rate, and repeat-purchase rate. Review data-quality exceptions alongside these metrics so business decisions are not based on unreliable records.

Important limitations

Delivered order value is item price plus freight for orders currently marked delivered; it is not net revenue or profit. Weekly groupings use purchase dates and current order status. The repeat-purchase rate is an overall dataset measure, not a weekly rate. The observed relationship between delays and reviews does not establish causation.